



Cloud Computing Lab Project

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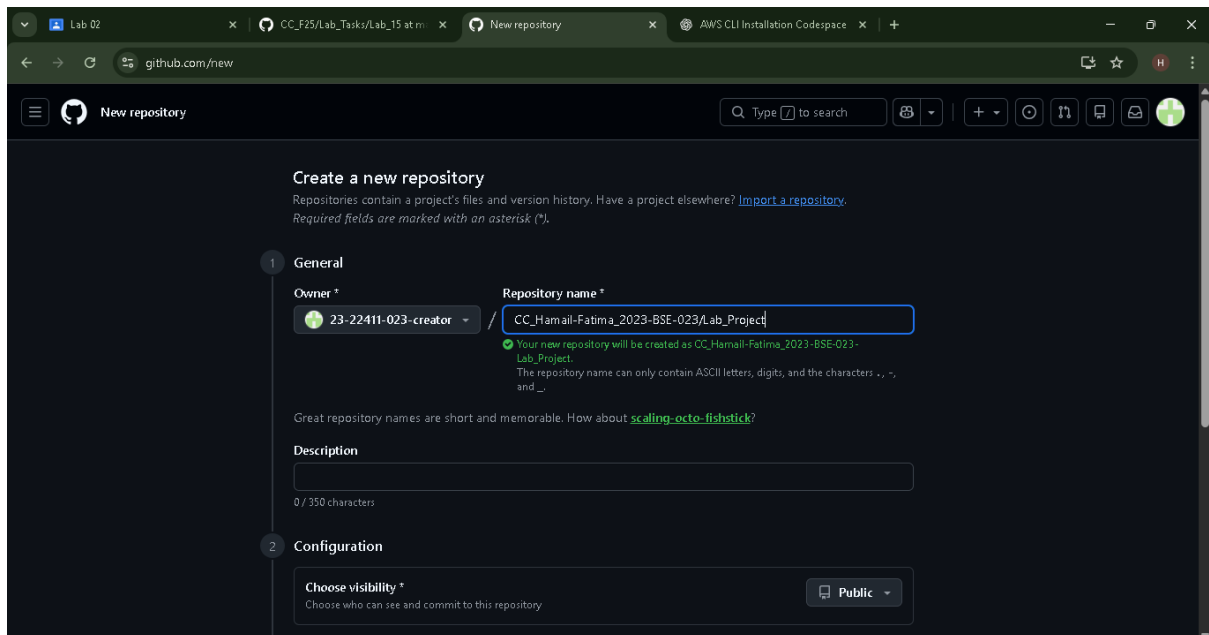
Roll No: 2023-BSE-023

Section: 5 A

Lab Project Repo: https://github.com/23-22411-023-creator/CC_Hamail-Fatima_2023-BSE-023-Lab_Project

Lab Project

STEP 1: Verify Prerequisites



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner * 23-22411-023-creator / Repository name * CC_Hamail-Fatima_2023-BSE-023/Lab_Project

✓ Your new repository will be created as CC_Hamail-Fatima_2023-BSE-023-Lab_Project.
The repository name can only contain ASCII letters, digits, and the characters -, ., and _.

Great repository names are short and memorable. How about [scaling-octo-fishstick](#)?

Description

0 / 350 characters

2 Configuration

Choose visibility * Public

Choose who can see and commit to this repository

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ tree
.
├── hosts.tpl
├── playbooks
├── site.yaml
├── roles
│   ├── backend
│   │   ├── handlers
│   │   │   └── main.yml
│   │   ├── tasks
│   │   │   └── main.yml
│   │   └── templates
│   │       └── backend_index.html.j2
│   ├── common
│   │   └── tasks
│   │       └── main.yml
│   └── frontend
│       ├── handlers
│       │   └── main.yml
│       ├── tasks
│       │   └── main.yml
│       └── templates
│           └── nginx_frontend.conf.j2
├── locals.tf
├── main.tf
├── modules
│   ├── subnet
│   └── webserver
├── outputs.tf
├── screenshots
├── terraform.tfvars
└── variables.tf
```

Verify AWS credentials

```

● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ aws configure
AWS Access Key ID [None]: AKIA42E3QACSNIPKW2KQG
AWS Secret Access Key [None]: 0V+BGmfMvkckRshZ6ztGZRIoMVqJXKeDf1/qA3V9
Default region name [None]: me-central-1
Default output format [None]: json
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ aws sts
get-caller-identity
{
  "UserId": "AIDA42E3QACSNIPKW2KQG",
  "Account": "880794402980",
  "Arn": "arn:aws:iam::880794402980:user/lab-user"
}
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $

```

Generate SSH key (if needed)

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ ssh-keygen
Generating public/private ed25519 key pair.
Created directory '/home/codespace/.ssh'.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:BFsqX3WGeMqH3114Vc/fTuO8PARdmEOhLrKQPjtgyR8 codespace@codespaces-aece9b
The key's randomart image is:
+--[ED25519 256]--+
|  . . . . 0.0+0 |
|    =.00 .+.+ |
|  . 0.0+ . .0+ |
|    0 ++ .. .+ = |
|  . .+ So.....+ |
|    =.E. 0... ++. |
|  . 00.. .0. |
|    00 ... |
|    .. 0. |
+-----[SHA256]-----+
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $

```

Verify tools

terraform version

ansible --version

python3 --version

```

● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform
version
ansible --version
python3 --version
Terraform v1.14.3
on linux_amd64
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Nov  6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
Python 3.12.1
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $

```

Step 2: Deploy Infrastructure

Navigate to project

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ cd /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Initialize Terraform

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform init
- Installed hashicorp/null v3.2.4 (signed by HashiCorp)
- Installing hashicorp/local v2.6.1...
- Installed hashicorp/local v2.6.1 (signed by HashiCorp)
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Deploy (creates all resources + runs Ansible)

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ cd /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project && terraform validate
Success! The configuration is valid.

● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ aws sts get-call
er-identity 2>&1 || echo "AWS credentials not configured - will need to configure"
{
  "UserId": "AIDA42E3QACSNIPKW2KQG",
  "Account": "880794402980",
  "Arn": "arn:aws:iam::880794402980:user/lab-user"
}

● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ cd /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project && terraform init 2>&1 | tail -20
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/null from the dependency lock file
- Reusing previous version of hashicorp/local from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
```

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ cd /workspaces/C
C_Hamail-Fatima_2023-BSE-023-Lab_Project && terraform init 2>&1 | tail -20
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ [ -f ~/.ssh/id_e
d25519 ] && echo "SSH key already exists" || (ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N "" && echo "
SSH key generated") && ls -lh ~/.ssh/id_ed25519*
SSH key already exists
-rw----- 1 codespace codespace 419 Jan 26 13:57 /home/codespace/.ssh/id_ed25519
-rw-r--r-- 1 codespace codespace 109 Jan 26 13:57 /home/codespace/.ssh/id_ed25519.pub
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ cd /workspaces/C
C_Hamail-Fatima_2023-BSE-023-Lab_Project && terraform plan -out=tfplan 2>&1 | tail -50
}
```

Plan: 14 to add, 0 to change, 0 to destroy.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform apply -
auto-approve
    "private_ip" = "10.0.1.160"
    "public_ip"  = "100.53.58.100"
  }
}
backend_private_ips = [
  "10.0.1.232",
  "10.0.1.125",
  "10.0.1.160",
]
backend_public_ips = [
  "3.231.22.33",
  "13.220.38.183",
  "100.53.58.100",
]
frontend_private_ip = "10.0.1.158"
frontend_public_ip  = "100.48.191.144"
security_group_id    = "sg-0f3e3f348bcda9699"
subnet_id            = "subnet-03adff558b139c1ef"
vpc_id               = "vpc-03510d4be6f9fa096"
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 3: Verify Deployment

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform output
"private_ip" = "10.0.1.232"
"public_ip" = "3.231.22.33"
}
"Backend-1" = {
  "id" = "i-07403b44486489947"
  "private_ip" = "10.0.1.125"
  "public_ip" = "13.220.38.183"
}
"Backend-2" = {
  "id" = "i-0a8fd0f752565bd79"
  "private_ip" = "10.0.1.160"
  "public_ip" = "100.53.58.100"
}
}
backend_private_ips = [
  "10.0.1.232",
  "10.0.1.125",
  "10.0.1.160",
]
backend_public_ips = [
  "3.231.22.33",
  "13.220.38.183",
  "100.53.58.100",
]
frontend_private_ip = "10.0.1.158"
frontend_public_ip = "100.48.191.144"
security_group_id = "sg-0f3e3f348bcd9699"
subnet_id = "subnet-03adff558b139c1ef"
vpc_id = "vpc-03510d4be6f9fa096"
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Test frontend load balancer

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform output
-raw frontend_public_ip
100.48.191.144@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Test individual backends

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform output
-json backend_private_ips | jq -r '.[0]' | jq -r '.[0]'
10.0.1.232
10.0.1.125
10.0.1.160
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 4: Test Load Balancing

Multiple requests should alternate between 2 primary backends

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform output
-json backend_private_ips | jq -r '.[0]'
terraform output -json backend_public_ips | jq -r '.[0]'
10.0.1.232
10.0.1.125
10.0.1.160
3.231.22.33
13.220.38.183
100.53.58.100
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Phase 4: Verify AWS Resources

Step 4.1: List EC2 Instances

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ aws ec2 describe
-instances --query 'Reservations[].Instances[].[InstanceId,State.Name,PublicIpAddress,PrivateIpAddress]'
--output table
```

DescribeInstances				
InstanceId	State.Name	PublicIpAddress	PrivateIpAddress	
i-0e2855fc9ce3da8cf	running	3.28.194.208	172.31.7.174	
i-0725e1c39d5916ea9	running	3.29.232.129	10.0.10.158	
i-07b139994b0afcda9	running	158.252.33.128	10.0.1.6	
i-0b759e2fe897099bc	running	3.28.184.123	10.0.1.203	
i-0cd53f08a0d6215d3	running	3.29.27.129	10.0.1.110	
i-0ed8548845dc16744	running	3.29.67.156	10.0.1.195	

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 4.2: List Security Groups

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ aws ec2 describe
-instances --query 'Reservations[].Instances[].[InstanceId,InstanceType,State.Name,PublicIpAddress,PrivateIpAddress]' --output table
```

DescribeInstances				
InstanceId	InstanceType	State.Name	PublicIpAddress	PrivateIpAddress
i-0e2855fc9ce3da8cf	t3.micro	running	3.28.194.208	172.31.7.174
i-0725e1c39d5916ea9	t3.micro	running	3.29.232.129	10.0.10.158
i-07b139994b0afcda9	t3.micro	running	158.252.33.128	10.0.1.6
i-0b759e2fe897099bc	t3.micro	running	3.28.184.123	10.0.1.203
i-0cd53f08a0d6215d3	t3.micro	running	3.29.27.129	10.0.1.110
i-0ed8548845dc16744	t3.micro	running	3.29.67.156	10.0.1.195

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 4.3: Verify VPC and Subnet

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ aws ec2 describe
-vpcs --query 'Vpcs[].[VpcID:VpcID,CidrBlock:CidrBlock,Name:Tags[?Key==`Name`].Value|[0]]' --output table
```

DescribeVpcs		
CidrBlock	Name	VpcID
10.0.0.0/16	lab-vpc	vpc-056c83bd182d43f12
172.31.0.0/16	None	vpc-02e5f1f03e6b88a38
10.0.0.0/16	dev-vpc	vpc-0c6f53becf083da95
10.0.0.0/16	lab-vpc	vpc-021e07ab2e6b265cc
10.0.0.0/16	development	vpc-0a2ccf4baa16a7c9b

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Phase 5: Verify Ansible Configuration

Step 5.1: Check Ansible Inventory

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ cat ansible/inventory/hosts
```

```
[frontend]
100.48.191.144

[backends]
3.231.22.33
13.220.38.183
100.53.58.100

[all:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_python_interpreter=/usr/bin/python3
```

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 5.2: Verify Ansible Connectivity

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ ansible all -i ansible/inventory/hosts -m ping
3.231.22.33 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
100.48.191.144 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
100.53.58.100 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
13.220.38.183 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 5.3: Gather Facts from All Hosts

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ ansible all -i ansible/inventory/hosts -m setup | head -100
    "uuids": {
      "xvda1": [
        "602aea34-1b1c-49c6-8d1f-d7de09dce53c"
      ]
    },
    "ansible_devices": {
      "xvda": {
        "holders": [],
        "host": "",
        "links": {
          "ids": [],
          "labels": [],
          "masters": [],
          "uuids": []
        },
        "model": null,
        "partitions": {
          "xvda1": {
            "holders": [],
            "links": {
[WARNING]: Failure using method (v2_runner_on_ok) in callback plugin
(<ansible.plugins.callback.minimal.CallbackModule object at 0x76f4b3f76060>):
[Errno 32] Broken pipe
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 5.4: Check Ansible Playbook Syntax

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project/ansible (main) $ ansible-playbook --syntax-check playbooks/site.yaml
playbook: playbooks/site.yaml
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project/ansible (main) $
```

Phase 6: Test Ansible Roles

Step 6.1: Run Ansible Playbook Manually


```

@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project/ansible (main) $ ansible-playbook \
-i inventory/hosts \
--private-key ~/.ssh/id_ed25519 \
-u ec2-user \ ...
PLAY [Gather facts from all hosts] *****

TASK [Gathering Facts] *****
ok: [3.231.22.33]
ok: [100.48.191.144]
ok: [13.220.38.183]
ok: [100.53.58.100]

TASK [Set backend IPs] *****
ok: [100.48.191.144 -> localhost] => {"ansible_facts": {"backend1_private_ip": "3.231.22.33", "backend2_private_ip": "13.220.38.183", "backup_backend_private_ip": "100.53.58.100"}, "changed": false}

PLAY [Configure backend HTTPD servers] *****

TASK [Gathering Facts] *****
ok: [3.231.22.33]
ok: [13.220.38.183]
ok: [100.53.58.100]

TASK [backend : Install Apache HTTPD] *****
ok: [3.231.22.33] => {"changed": false, "msg": "", "rc": 0, "results": ["httpd-2.4.66-1.amzn2.0.1.x86_64 providing httpd is already installed"]}

```

Step 6.2: Check Backend HTTPD Status

```

@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ BACKEND1=$(terraform
orm output -json backend_public_ips | jq -r '.[0]')
ssh -i ~/.ssh/id_ed25519 ec2-user@$BACKEND1 "sudo systemctl status httpd"
Active: active (running) since Mon 2026-01-26 15:11:36 UTC; 2min 28s ago
Docs: man:httpd.service(8)
Main PID: 7544 (httpd)
Status: "Total requests: 1; Idle/Busy workers 100/0;Requests/sec: 0.00719; Bytes served/sec: 13 B/sec"
c"
CGroup: /system.slice/httpd.service
├─7544 /usr/sbin/httpd -DFOREGROUND
├─7546 /usr/sbin/httpd -DFOREGROUND
├─7547 /usr/sbin/httpd -DFOREGROUND
├─7548 /usr/sbin/httpd -DFOREGROUND
├─7549 /usr/sbin/httpd -DFOREGROUND
├─7550 /usr/sbin/httpd -DFOREGROUND
└─7602 /usr/sbin/httpd -DFOREGROUND

Jan 26 15:11:36 ip-10-0-1-232.ec2.internal systemd[1]: Stopped The Apache HTTP Server.
Jan 26 15:11:36 ip-10-0-1-232.ec2.internal systemd[1]: Starting The Apache HTTP Server...
Jan 26 15:11:36 ip-10-0-1-232.ec2.internal systemd[1]: Started The Apache HTTP Server.
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $

```

Step 6.3: Check Frontend Nginx Status

```

@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ FRONTEND_IP=$(terraform output -raw frontend_public_ip)
ssh -i ~/.ssh/id_ed25519 ec2-user@$FRONTEND_IP "sudo systemctl status nginx"
• nginx.service - The nginx HTTP and reverse proxy server
  Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; vendor preset: disabled)
  Active: active (running) since Mon 2026-01-26 14:33:12 UTC; 41min ago
  Process: 5524 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
  Process: 5520 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
  Process: 5518 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
  Main PID: 5526 (nginx)
  CGroup: /system.slice/nginx.service
          └─5526 nginx: master process /usr/sbin/nginx
            └─5527 nginx: worker process

Jan 26 14:33:12 ip-10-0-1-158.ec2.internal systemd[1]: Stopped The nginx HTTP and reverse proxy server.
Jan 26 14:33:12 ip-10-0-1-158.ec2.internal systemd[1]: Starting The nginx HTTP and reverse proxy server.
..
Jan 26 14:33:12 ip-10-0-1-158.ec2.internal nginx[5520]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jan 26 14:33:12 ip-10-0-1-158.ec2.internal nginx[5520]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jan 26 14:33:12 ip-10-0-1-158.ec2.internal systemd[1]: Started The nginx HTTP and reverse proxy server.
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ █

```

Phase 7: Test Load Balancing

Step 7.1: Access Individual Backend Servers

```

@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform output -json backend_public_ips | jq -r '.[0]' | while read ip; do
  echo "Backend $ip:"
  curl -s http://$ip/ | grep -A 2 "Backend server"
  echo "---"
done
Backend 3.231.22.33:
---
Backend 13.220.38.183:
---
Backend 100.53.58.100:
---
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ █

```

Step 7.2: Test Frontend Load Balancer

```

@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ FRONTEND_IP=$(terraform output -raw frontend_public_ip)
for i in {1..6}; do
  echo "Request $i:"
  curl -s http://$FRONTEND_IP/ | grep -A 1 "Backend server"
  sleep 1
done
Request 1:
Request 2:
Request 3:
Request 4:
Request 5:
Request 6:
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ █

```

Step 7.3: View Nginx Logs

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ ssh -i ~/.ssh/id_
ed25519 ec2-user@$FRONTEND_IP \
  "sudo tail -20 /var/log/nginx/access.log"
4.240.18.230 - - [26/Jan/2026:14:36:10 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:14:36:29 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:14:36:30 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:14:36:30 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:14:36:31 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:14:36:44 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:14:39:45 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
103.149.241.60 - - [26/Jan/2026:14:51:15 +0000] "GET / HTTP/1.1" 200 1525 "-" "Mozilla/5.0 (Windows NT 1
0.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
103.149.241.60 - - [26/Jan/2026:14:51:15 +0000] "GET /favicon.ico HTTP/1.1" 404 236 "http://100.48.191.1
44/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0
Safari/537.36" "-"
204.76.203.219 - - [26/Jan/2026:15:03:12 +0000] "GET / HTTP/1.1" 200 1525 "-" "Mozilla/5.0 (Windows NT 1
0.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/90.0.4430.85 Safari/537.36 Edg/90.0.818.4
6" "-"
4.240.18.230 - - [26/Jan/2026:15:17:39 +0000] "GET / HTTP/1.1" 200 1521 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:15:17:40 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:15:17:42 +0000] "GET / HTTP/1.1" 200 1521 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:15:17:43 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:15:17:45 +0000] "GET / HTTP/1.1" 200 1521 "-" "curl/8.5.0" "-"
4.240.18.230 - - [26/Jan/2026:15:17:46 +0000] "GET / HTTP/1.1" 200 1525 "-" "curl/8.5.0" "-"
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Phase 8: Test Failover Behavior

Step 8.1: Stop Primary Backend 1

```
• @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ BACKEND1=$(terraf
orm output -json backend_public_ips | jq -r '[0]')
ssh -i ~/.ssh/id_ed25519 ec2-user@$BACKEND1 \
  "sudo systemctl stop httpd"
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Verify it's stopped

```
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ ssh -i ~/.ssh/id_
ed25519 ec2-user@$BACKEND1 \
  "sudo systemctl status httpd"
• httpd.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
  Active: inactive (dead) since Mon 2026-01-26 15:19:25 UTC; 32s ago
  Docs: man:httpd.service(8)
  Process: 7544 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, status=0/SUCCESS)
  Main PID: 7544 (code=exited, status=0/SUCCESS)
  Status: "Total requests: 5; Idle/Busy workers 100/0; Requests/sec: 0.0109; Bytes served/sec: 21 B/sec"

Jan 26 15:11:36 ip-10-0-1-232.ec2.internal systemd[1]: Stopped The Apache HTTP Server.
Jan 26 15:11:36 ip-10-0-1-232.ec2.internal systemd[1]: Starting The Apache HTTP Server...
Jan 26 15:11:36 ip-10-0-1-232.ec2.internal systemd[1]: Started The Apache HTTP Server.
Jan 26 15:19:24 ip-10-0-1-232.ec2.internal systemd[1]: Stopping The Apache HTTP Server...
Jan 26 15:19:25 ip-10-0-1-232.ec2.internal systemd[1]: Stopped The Apache HTTP Server.
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 8.2: Test Frontend Access During Failover

```
• @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ FRONTEND_IP=$(ter
raform output -raw frontend_public_ip)
for i in {1..5}; do
  echo "Request $i (Backend1 stopped):"
  curl -s http://$FRONTEND_IP/ | grep -A 1 "Backend server"
  sleep 1
done
Request 1 (Backend1 stopped):
Request 2 (Backend1 stopped):
Request 3 (Backend1 stopped):
Request 4 (Backend1 stopped):
Request 5 (Backend1 stopped):
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 8.3: Stop Primary Backend 2

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ BACKEND2=$(terraform output -json backend_public_ips | jq -r '.[1]')
ssh -i ~/.ssh/id_ed25519 ec2-user@$BACKEND2 \
  "sudo systemctl stop httpd"
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 8.4: Test Backup Backend Serving Requests

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ FRONTEND_IP=$(terraform output -raw frontend_public_ip)
for i in {1..5}; do
  echo "Request $i (Both primaries stopped):"
  curl -s http://$FRONTEND_IP/ | grep -A 1 "Backend server"
  sleep 1
done
Request 1 (Both primaries stopped):
Request 2 (Both primaries stopped):
Request 3 (Both primaries stopped):
Request 4 (Both primaries stopped):
Request 5 (Both primaries stopped):
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Step 8.5: Restart Both Backends

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ BACKEND1=$(terraform output -json backend_public_ips | jq -r '.[0]')
BACKEND2=$(terraform output -json backend_public_ips | jq -r '.[1]')

ssh -i ~/.ssh/id_ed25519 ec2-user@$BACKEND1 "sudo systemctl start httpd"
ssh -i ~/.ssh/id_ed25519 ec2-user@$BACKEND2 "sudo systemctl start httpd"
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Verify

```
● @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ FRONTEND_IP=$(terraform output -raw frontend_public_ip)
for i in {1..6}; do
  echo "Request $i (All backends running):"
  curl -s http://$FRONTEND_IP/ | grep -A 1 "Backend server"
  sleep 1
done
Request 1 (All backends running):
Request 2 (All backends running):
Request 3 (All backends running):
Request 4 (All backends running):
Request 5 (All backends running):
Request 6 (All backends running):
○ @23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

Phase 9: Verify Nginx Configuration

Step 9.1: Check Nginx Config on Frontend

```

@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ ssh -i ~/.ssh/id_
ed25519 ec2-user@$FRONTEND_IP \
"sudo cat /etc/nginx/nginx.conf | grep -A 20 'upstream backend'"
server 3.231.22.33:80 max_fails=3 fail_timeout=30s;
server 13.220.38.183:80 max_fails=3 fail_timeout=30s;

# Backup backend (used only when primaries are down)
server 100.53.58.100:80 backup max_fails=3 fail_timeout=30s;
}

# ===== SERVER CONFIGURATION =====
server {
    listen 80;
    server_name _;

    # Redirect HTTP to HTTPS if needed (optional)
    # return 301 https://$server_name$request_uri;

    # ===== ROOT LOCATION =====
    location / {
        proxy_pass http://backend_servers;
        proxy_http_version 1.1;

```

Step 9.2: Test Nginx Configuration

```

@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ ssh -i ~/.ssh/id_
ed25519 ec2-user@$FRONTEND_IP \
"sudo nginx -t"
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $

```

Not secure
100.48.191.144

Backend Server

Hostname: 13.220.38.183

Private IP: 10.0.1.125

Node Name: ip-10-0-1-125

OS: Amazon 2

Server Role: HTTPD Backend Server

Request Timestamp: 2026-01-26T14:31:31Z

Backend Server

Hostname: 3.231.22.33

Private IP: 10.0.1.232

Node Name: ip-10-0-1-232

OS: Amazon 2

Server Role: HTTPD Backend Server

Request Timestamp: 2026-01-26T15:10:54Z

Phase 10: Idempotence Testing

Step 10.1: Re-run Terraform Apply

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $ terraform apply -
auto-approve
  "10.0.1.232",
  "10.0.1.125",
  "10.0.1.160",
]
backend_public_ips = [
  "3.231.22.33",
  "13.220.38.183",
  "100.53.58.100",
]
frontend_private_ip = "10.0.1.158"
frontend_public_ip = "100.48.191.144"
security_group_id = "sg-0f3e3f348bcd9699"
subnet_id = "subnet-03adff558b139c1ef"
vpc_id = "vpc-03510d4be6f9fa096"
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project (main) $
```

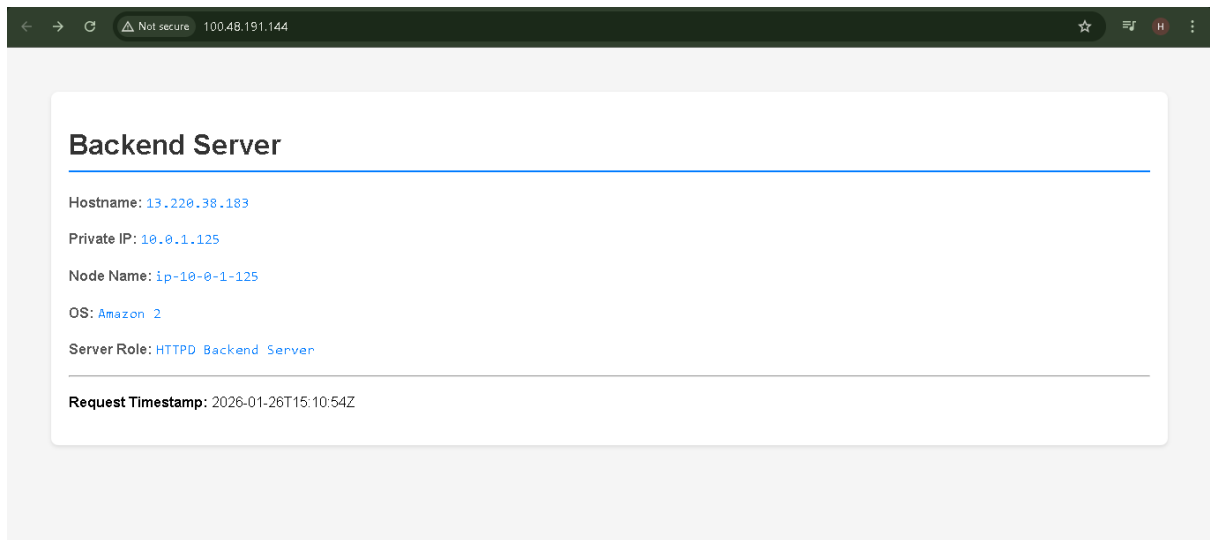
Step 10.2: Re-run Ansible Playbook

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
@23-22411-023-creator → /workspaces/CC_Hamail-Fatima_2023-BSE-023-Lab_Project/ansible (main) $ ansible-p
laybook \
-i inventory/hosts \
--private-key ~/.ssh/id_ed25519 \
-u ec2-user \ ...
guration file /etc/nginx/nginx.conf test is successful", "stderr_lines": ["nginx: the configuration file
/etc/nginx/nginx.conf syntax is ok", "nginx: configuration file /etc/nginx/nginx.conf test is successfu
l"], "stdout": "", "stdout_lines": []}

PLAY RECAP *****
100.48.191.144      : ok=9    changed=2    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
100.53.58.100      : ok=8    changed=2    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
13.220.38.183      : ok=8    changed=2    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
3.231.22.33        : ok=8    changed=3    unreachable=0    failed=0    skipped=0    rescued=0
```

Phase 11:

Step 11.1 : Front End with Backend 1



Step 11.2: Front End with Backend 2

